

Single photons on demand from a single molecule at room temperature

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The generation of non-classical states of light¹ is of fundamental scientific and technological interest. For example, 'squeezed' states² enable measurements to be performed at lower noise levels than possible using classical light. Deterministic (or triggered) single-photon sources exhibit non-classical behaviour in that they emit, with a high degree of certainty, just one photon at a user-specified time. (In contrast, a classical source such as an attenuated pulsed laser emits photons according to Poisson statistics.) A deterministic source of single photons could find applications in quantum information processing³, quantum cryptography⁴ and certain quantum computation problems⁵. Here we realize a controllable source of single photons using optical pumping of a single molecule in a solid. Triggered single photons are produced at a high rate, whereas the probability of simultaneous emission of two photons is nearly zero—a useful property for secure quantum cryptography. Our approach is

characterized by simplicity, room temperature operation and improved performance compared to other triggered sources of single photons.

Various schemes have been proposed to create a single-photon source, for example, involving single atoms in cavities^{6–8}, or highly nonlinear cavities⁹. Until now, such sources have only been demonstrated in two experiments at cryogenic temperatures. The first such demonstration utilized a "turnstile" effect¹⁰ based on a Coulomb blockade for electrons and holes in a mesoscopic double-barrier p–n junction¹¹. A dilution refrigerator operating at 50 mK was essential to minimize the thermal energy background. Owing to the sample geometry, the detection efficiency was limited to about 10^{-4} , which made it difficult to measure the second-order intensity correlation of the emitted photons, an important diagnostic of the non-classical nature of the light.

A second approach is based on the controlled excitation of single molecules in a solid¹². A rapid adiabatic passage method is used to prepare the molecule in its fluorescent state. Very narrow optical absorption lines are needed for this method, which requires liquid helium temperatures (2 K). The detection efficiency was limited to about 3×10^{-3} , mainly because filtering out the pumping laser excitation also eliminates single-molecule fluorescence near the laser frequency.

Our source of single photons on demand is based on a simple concept, the pulsed optical excitation of a single highly fluorescent molecule as illustrated in Fig. 1a. A short pulse of green laser light pumps the four-level scheme of the molecule from the ground singlet state to a vibronically excited level of the first electronic excited singlet state. After fast (picoseconds) relaxation to the lowest electronic excited state, the molecule subsequently emits a single photon. Because the molecule can be pumped at high intensity, the probability of preparing the emitting state can approach unity. This scheme has the further advantage of spectrally separating the laser excitation and the fluorescence emission.

Fluorescence microscopy at ambient conditions of single emissive dye molecules^{13,14} embedded in polymers, droplets and gels or dispersed on surfaces has revealed various previously unknown effects that are normally obscured by averaging in conventional ensemble measurements. Digital photobleaching¹⁵ is an example; however, it imposes an upper limit on the number of photons detectable from a molecule. The photobleaching quantum efficiency of most single molecules has been reported to range from about 10^{-4} to about 10^{-7} in various host matrices, which is a severe limitation for quantum optical experiments^{16–18}. However,

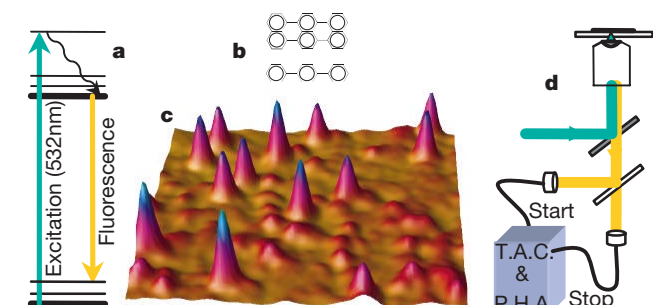


Figure 1 Scheme of the experimental measurements with images of single molecules. **a**, Pumping scheme of the molecules to their fluorescent state. **b**, Chemical structures of terrylene (upper) and p-terphenyl (lower). The sample is a sublimed crystal flake (a few micrometres in thickness) of p-terphenyl doped with terrylene at low concentrations of about 10^{-11} moles of terrylene per mole of p-terphenyl. Terrylene fluoresces at around 579 nm in the p-terphenyl crystal, and has a fluorescence quantum yield close to unity. **c**, Confocal fluorescence image ($10 \mu\text{m} \times 10 \mu\text{m}$) of single terrylene molecules embedded in crystalline p-terphenyl with continuous-wave excitation at 532 nm of about $1.5 \mu\text{W}$, with signal-to-background ratio greater than 5. **d**, Schematic representation of the scanning confocal microscope set-up used to characterize the single-molecule emission. The laser excitation was provided by a continuous-wave 532 nm frequency-doubled Nd-YAG laser for the antibunching measurement. For the pulsed excitation we used an actively mode-locked picosecond Nd-YAG laser from Lightwave Electronics (model 131, pulse width 35 ps, repetition rate 100 MHz, average power 220 mW). A pulse picker was used to reduce the repetition rate to 6.25 MHz. The pulsed green light was produced by single pass second-harmonic generation in a periodically poled lithium niobate crystal²⁸ (a maximum average green power of 0.2 mW was obtained for an average near infrared power of 10 mW). The excitation beam was directed onto the sample by the dichroic mirror of an inverted microscope and was focused on the sample plane by a 1.4 NA (numerical aperture) oil-immersion objective. The fluorescence photons were collected by this objective, and filtered from the residual excitation light by a holographic notch filter and long pass glass filter. The emitted fluorescence was sent through a 50/50 beam splitter onto two (start and stop) Si avalanche photon-counting photodiodes. The histogram of the delays between detected photons was measured using a time-to-amplitude converter and a pulse-height analyser. The stop signal was delayed for negative times.

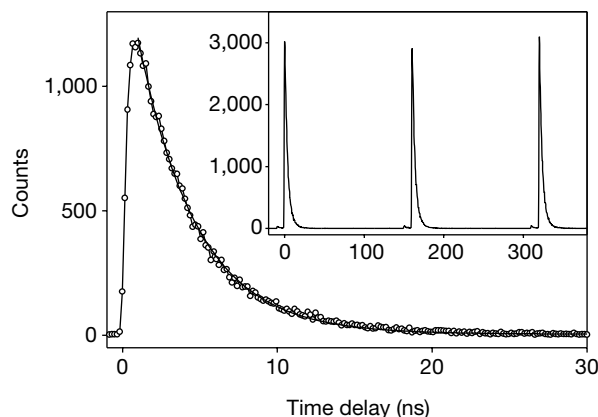


Figure 2 Characterization of the average time structure of the source. The arrival times of fluorescence photons are recorded with one detector using a time-to-amplitude converter triggered by the laser pulse. The average power of the pulsed laser is $6 \mu\text{W}$ and the acquisition time is 60 s. Circles represent the histogram of arrival times of detected photons after the pump pulse at $t = 0$. The line is a single exponential decay fit (decay time of 3.8 ns). Inset, wide time range.

highly stable single terrylene molecules have recently been detected in p-terphenyl molecular crystals at room temperature^{19,20} (for structures see Fig. 1b). In this organic crystal, the fluorescent molecules are protected by the host from exposure to diffusing quenchers (such as oxygen), and profit from the ability to emit host phonons to prevent thermally induced damage. We find that for thick crystals ($\sim 10 \mu\text{m}$), this system indeed has extremely high photostability, allowing hours of continuous illumination without photobleaching.

Figure 1c shows a scanning confocal microscope image of the fluorescence from single terrylene molecules embedded in a thin, sublimed platelet of crystalline p-terphenyl. The isolated peaks ($\sim 400 \text{ nm}$ full-width at half-maximum) represent the emission of single molecules. Saturation studies that we performed on this system show that maximum emission rates as high as 2.5 MHz are achievable, and typical saturation intensities are close to 1 MW cm^{-2} . The high saturation intensities are the result of the orientation of the molecular transition dipole moment which is nearly perpendicular to the plane of the sample and hence to the laser polarization.

The light emitted by a single molecule excited by a continuous-wave laser consists of single photons separated by random time intervals which depend on the excited state lifetime and the pumping rate. Because a single molecule can never emit two photons at once, the distribution of photon pairs separated by a time τ rises from zero at $\tau = 0$. This phenomenon, called photon antibunching, has previously been observed for single molecules at low²¹ and room temperature²². Using the standard Hanbury-Brown and Twiss coincidence set-up as shown in Fig. 1d, we observed strong antibunching correlations (data not shown), an unequivocal signature of the single-molecule nature of the peaks shown in Fig. 1c.

To use the non-classical fluorescence properties of the single molecule directly, we trigger the single-photon emission by pumping with a periodic, pulsed laser source (pulse width $\tau_p = 35 \text{ ps}$, repetition rate $\nu = 6.25 \text{ MHz}$). Single photons are then generated at predetermined times, within the accuracy of the emission lifetime of a few nanoseconds. We used the standard time-correlated single-photon counting technique to measure the average time structure of our source (see Fig. 2). The inset shows that the photons are emitted at times separated by the laser repetition time (160 ns). As expected, the main figure shows that the excited state population rises in a very short time, demonstrating the rapid pumping of the molecule to its emitting state. The distribution then decays exponentially with a time constant of $\tau_f = 3.8 \text{ ns}$, consistent with the lifetime of terrylene in p-terphenyl as deduced from the homogeneous width of approximately 40 MHz at low temperature²³.

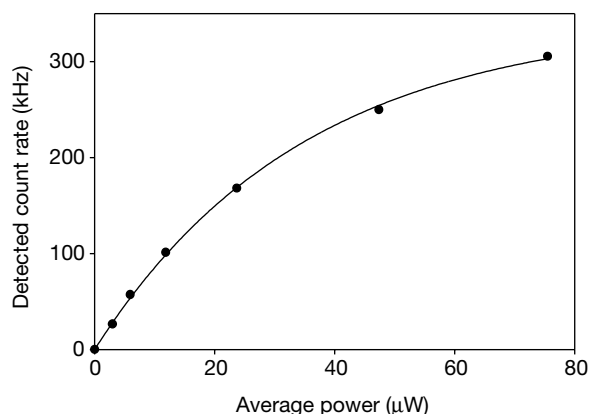


Figure 3 Determination of $p(1)$. The figure shows the intensity dependence of the emission rate from the triggered single photon source (dots) with a two-level model fit to the saturation behaviour (solid line). The maximum measured count rate is 310 kHz, and the expected saturation count rate from the fit is $S_\infty = 343 \text{ kHz}$, with $I_s = 1.2 \text{ MW cm}^{-2}$.

We now discuss the probabilities $p(m)$ that m photons will be emitted by the molecule after each pulse. Since the pump pulse width is very short compared to the fluorescence lifetime, the probability that two photons will be emitted per pulse, $p(2)$, is very small (see below). Thus the detected count rate from the molecule, S , is directly proportional to the probability $p(1)$ that a single photon is emitted after each laser pulse: $S = \eta \nu p(1)$ where η is the detection efficiency. By consideration of the losses in filters and optics as well as the collection efficiency for a molecular dipole oriented nearly perpendicular to the plane of the microscope stage²⁴, we find $\eta \approx 6\%$. To estimate the maximum value of $p(1)$, we performed a power saturation study of the detected count rate. In Fig. 3 we show the measured S as a function of the average excitation power of the pulsed laser. The signal shows power saturation behaviour that is well fitted by the saturation law of a two-level system from rate equations, $S = S_\infty (I/I_s) / (1 + I/I_s) [1 - \exp\{-(1 + I/I_s)\tau_p/\tau_f\}] = S_\infty p(1)$ where I and I_s are respectively the excitation and the saturation intensities. From the fit we obtain a saturation count rate $S_\infty = 343 \pm 6 \text{ kHz}$ and $I_s = 1.2 \pm 0.1 \text{ MW cm}^{-2}$ (in agreement with continuous-wave measurements, not shown). The maximum measured count rate in our experiment reached 310 kHz and was limited only by the available laser power. The maximum achieved $p(1)$ can be determined in two ways: using our maximum pumping intensity and I_s determined above, $p(1) = 88\%$; using the maximum observed S and η , $p(1) = 83\%$, in good agreement.

A small value of double-photon emission, $p(2)$, is essential to the usefulness of a single-photon source. For example, quantum key distribution using polarization states of photons requires immunity from a beam-splitter attack²⁵, where a large value of $p(2)$ allows an intruder to obtain information without being detected. Two-photon emission requires that the molecule must first be excited, then a photon must be emitted, and then the molecule must be

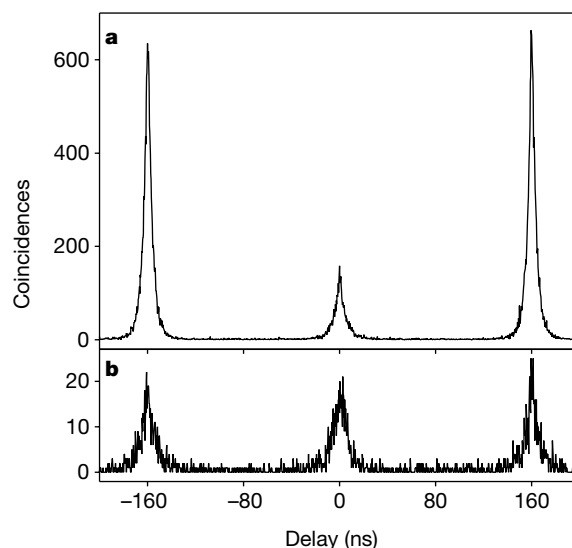


Figure 4 Second-order intensity correlation function of the emitted photons. The traces show the histogram of delays between detected photons under pulsed excitation with $6 \mu\text{W}$ average power measured with the start-stop apparatus of Fig. 1d. **a**, A single molecule is excited and the ratio of the area of the central peak to the lateral ones is 0.27. The area of the central peak is proportional to $(B^2 + 2SB)$, where B is the average rate detected from the background. In addition to the background-background events, a larger contribution arises from the signal-background and background-signal events for the start-stop coincidences. Similarly, the area of each lateral peak is proportional to $(B + S^2)$. The ratio of the area of the central peak to the lateral peak is in good quantitative agreement with the ratio expected from the measured average count rate for the signal and the background ($S/B \approx 6$). **b**, When the laser spot is positioned away from any molecule, all peaks have the same area, as expected for a poissonian source. The data were accumulated for **a**, 120 s and **b**, 300 s.

excited a second time, all within 35 ps. We have estimated the value of $p(2)$ and find it to be less than 8×10^{-4} at our maximum intensity.

To directly demonstrate the non-classical sub-poissonian statistics of our source, we measured the intensity correlation function of the emitted light with the coincidence set-up of Fig. 1d. For both the single-molecule (Fig. 4a) and background (Fig. 4b) cases, the histograms show the expected peak pattern, given the time pattern of the photon emission in the inset of Fig. 2. For poissonian light, such as that from an attenuated pulsed laser or the fluorescence from the background excited by the laser pulses, the central peak is identical in intensity and shape to the lateral ones (Fig. 4b). In the case of a single molecule, the central peak should vanish altogether as no more than one single photon can be emitted by the molecule. The ratio of the central peak's area to the area of the lateral peaks is the signature of the sub-poissonian statistics of the light emitted by our source. The residual peak at zero delay in Fig. 4a arises from coincidence events involving background photons excited during each laser pulse. In our experiment, the background signal shows a lifetime of about 4 ns, which indicates that it arises from weak fluorescence from out-of-focus terrylene molecules, not from Raman scattering.

It is convenient to compare the probability distribution $p(m)$ for our source to that expected from a Poisson distribution by means of the Mandel parameter $Q_s = (\sigma^2 - n_{av})/n_{av}$, where σ^2 is the variance of the distribution and n_{av} is the average number of photons²⁶. At the highest pumping power, the probabilities of our source are $p(0) = 0.14$, $p(1) = 0.86$ and $p(m > 1) \approx 0$. This distribution is radically different from that for a pulsed coherent source with the same $n_{av} = 0.86$: $p_{coh}(0) = 0.42$, $p_{coh}(1) = 0.36$, $p_{coh}(2) = 0.16$, The Mandel parameter of our source is $Q_s = -0.86$, not far from -1 , the value expected for a perfect single-photon emitter, and far from 0, the value for a poissonian source. The Mandel parameter Q_d of the detected photon counts is naturally affected by the light detection efficiency²⁷. Using $Q_d = Q_s/(\eta/2)$, we find $Q_d \approx -3\%$.

The parameters of our source (repetition rate and single-photon generation probability) are limited only by the laser system used; nevertheless, the current performance already surpasses that of previous work. This high performance combined with the simplicity of our source may make it suitable for a variety of quantum optical experiments and for other applications where triggered single photons are needed. Photons are emitted into a range of solid angles, which can limit the detection efficiency; however, optical solutions to this problem can be envisaged, that is, one can imagine that the single molecule could be coupled to a single cavity mode to reduce losses, change the emission pattern, or modify the emission lifetime and thus increase the emission rate. To reduce the background (from out-of-focus molecules or Raman scattering), reduced terrylene doping, pumping with z-axis polarized light, or use of a crystalline system with a more favourable orientation of the single absorber can be utilized. With further development, single molecules in solids may soon provide compact and reliable sources of single photons. □

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Logical computation using algorithmic self-assembly of DNA triple-crossover molecules

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Recent work^{1–3} has demonstrated the self-assembly of designed periodic two-dimensional arrays composed of DNA tiles, in which the intermolecular contacts are directed by 'sticky' ends. In a mathematical context, aperiodic mosaics may be formed by the self-assembly of 'Wang' tiles⁴, a process that emulates the operation of a Turing machine. Macroscopic self-assembly has been used to perform computations⁵; there is also a logical equivalence between DNA sticky ends and Wang tile edges^{6,7}. This suggests that the self-assembly of DNA-based tiles could be used to perform DNA-based computation⁸. Algorithmic aperiodic self-assembly requires greater fidelity than periodic self-assembly,